

Total No. of Questions: 8]

SEAT No. :

PA-993

[5927]-467

[Total No. of Pages : 2

B.E. (Computer Engineering) (Honours)
VIRTUAL REALITY IN GAME DEVELOPMENT
(2019 Pattern) (Semester-VII) (410701)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) Explain block diagram for a multimodal VR system developed at HRL Laboratories. [6]
b) Explain different types of project constraints in define stage. [6]
c) What is INVEST in user stories? [5]

OR

- Q2)** a) State different tools for building VR applications. [6]
b) Write short note on: [6]
i) Define Stage
ii) Make Stage
iii) Learn Stage
c) Virtual reality is both an art and a science. Justify. [5]
- Q3)** a) What are Fundamentals of sprite animation? [9]
b) Comparison between animations in Unreal Engine and Unity Engine. [9]

OR

- Q4)** a) Explain step to develop town view using unity. [9]
b) Explain working with Sprites in Unity. Also describe how to actually use Sprites in Unity. [9]
- Q5)** a) How to building in-game menu structures. Explain. [9]
b) Write unity code to prepare attack a single enemy in battle system. [9]

OR

P.T.O.

- Q6)** a) How to Implement the turn-based battle system. Explain. [9]
b) Explain workflow of Mecanim Animation System. [9]

- Q7)** a) Explain unified model of motion sickness with diagram. [6]
b) Write short note on: (Any 2). [6]
i) Binocular-Occlusion Conflict
ii) Flicker
iii) Aftereffects
c) State hardware design guidelines to avoid adverse health effects in VR System. [5]

OR

- Q8)** a) Explain different physical issues involved with the use of VR equipment. [6]
b) What are different application design factors that have adverse effects on VR system? [6]
c) What is Timing Analysis? Draw timing diagram for a typical VR system. [5]

